Drift-Sense - Final Reference Justification

References used to justify the physical effects included in the synthetic image generation pipeline.

DRAM STRUCTURE

D01 - IRDS 2024 - More Moore

Why it is used:
This reference provides the current semiconductor and DRAM scaling context. It highlights the importance of dimensional control and metrology as memory structures become smaller.

Source:
https://irds.ieee.org/images/files/pdf/2024/2024IRDS_MM.pdf

Main use in project:
Used to support the overall DRAM scaling and geometry context of the synthetic reference structures.

D02 - ITRS 2.0 - More Moore (2015)

Why it is used:
This reference covers semiconductor scaling, critical dimensions and process variation. It supports treating dimensional variation as a realistic part of nanoscale memory structures.

Source:
https://www.semiconductors.org/wp-content/uploads/2018/06/5_2015-ITRS-2.0_More-Moore.pdf

Main use in project:
Used to support the DRAM scaling, LER/CDU and metrology context used in the generator.

D03 - EP0967652A1 - DRAM cell array layout

Why it is used:
This patent provides a concrete example of a repeating DRAM cell-array structure with wordlines, bitlines and contacts. It was useful for grounding the synthetic geometry in a real memory-array layout.

Source:
https://patents.google.com/patent/EP0967652A1/en

Main use in project:
Used as a structural reference for the repeating DRAM geometry.

D04 - EP0780901A2 - DRAM memory-cell array

Why it is used:
This reference shows another practical DRAM memory-cell arrangement and its repeating layout. It supports using periodic structures rather than an arbitrary synthetic pattern.

Source:
https://patents.google.com/patent/EP0780901A2/en

Main use in project:
Used to support the periodic DRAM cell and array structure.

SEM NOISE

N01 - Sim, Thong & Phang - Effect of Shot Noise and Secondary Emission Noise in SEM Images (2004)

Why it is used:
This work discusses shot noise and secondary-electron emission noise in SEM images. It supports modelling SEM measurements as stochastic detector observations rather than ideal pixel values.

Source:
https://onlinelibrary.wiley.com/doi/10.1002/sca.4950260106

Main use in project:
Used to justify the signal-dependent shot-noise component in the synthetic SEM image.

N02 - Timischl, Date & Nemoto - A Statistical Model of Signal-Noise in SEM

Why it is used:
This reference provides a statistical treatment of SEM signal and measurement noise. It supports separating the underlying image signal from detector and readout variation.

Source:
https://www.researchgate.net/publication/51625528_A_statistical_model_of_signal-noise_in_scanning_electron_microscopy

Main use in project:
Used to support the detector/readout noise component of the generator.

N03 - Sakakibara et al. - Impact of Secondary Electron Emission Noise in SEM (2019)

Why it is used:
This work studies secondary-electron emission noise in semiconductor inspection SEM. It provides supporting evidence that detector-level stochastic variation is relevant to semiconductor SEM imaging.

Source:
https://pubmed.ncbi.nlm.nih.gov/30843582/

Main use in project:
Used as supporting justification for secondary-electron noise in the search image.

SEM PSF / BLUR

P01 - Kamal & Hailstone - Determination of the Optical PSF in the SEM Using Single Nanoparticle Images (2024)

Why it is used:
This work deals with measuring the point-spread function of an SEM imaging system. It supports the fact that SEM images have a finite spatial response rather than perfect point-to-point reproduction.

Source:
https://arxiv.org/abs/2407.01439

Main use in project:
Used to justify the finite PSF and blur applied during synthetic image formation.

P02 - Roels et al. - Bayesian Deconvolution of SEM Images Using PSF Estimation and Non-local Regularization (2018)

Why it is used:
This reference discusses SEM image formation, PSF estimation and image degradation. It provides direct support for modelling a finite imaging response together with measurement noise.

Source:
https://arxiv.org/abs/1810.09739

Main use in project:
Used for the PSF/blur and SEM noise formulation in the generator.

P03 - Postek et al. - Comparison of Electron Imaging Modes for Dimensional Measurements in the SEM

Why it is used:
This work examines SEM imaging behaviour for dimensional measurements and the effect of imaging conditions on observed features. It supports using realistic SEM contrast rather than an ideal rendered image.

Source:
https://pmc.ncbi.nlm.nih.gov/articles/PMC5113026/

Main use in project:
Used to support realistic feature contrast and imaging behaviour.

LER / CDU VARIATION

L01 - Bunday et al. - CD-SEM Measurement of LER Test Patterns

Why it is used:
This reference deals directly with measuring line-edge roughness using CD-SEM. It supports including controlled edge variation in semiconductor structures.

Source:
https://www.nist.gov/publications/cd-sem-measurement-line-edge-roughness-test-patterns-193-nm-lithography

Main use in project:
Used to justify the LER component of the synthetic structures.

L02 - Bunday et al. - Determination of Optimal Parameters for CD-SEM Measurement of LER

Why it is used:
This work focuses on the measurement of LER using CD-SEM and the parameters affecting that measurement. It supports treating edge roughness as a meaningful semiconductor metrology variation.

Source:
https://www.nist.gov/publications/determination-optimal-parameters-cd-sem-measurement-line-edge-roughness

Main use in project:
Used as supporting reference for LER and CD-SEM variation.

L03 - Constantoudis et al. - Line Edge Roughness and Critical Dimension Variation

Why it is used:
This reference examines the relationship between line-edge roughness and critical-dimension variation. It supports modelling edge variation together with small dimensional changes.

Source:
https://www.researchgate.net/publication/249512732_Line_edge_roughness_and_critical_dimension_variation

Main use in project:
Used to support the combined LER and CDU variation in the synthetic structures.

OVERLAY / REGISTRATION

O01 - Sullivan & Shin - Overlay Metrology: The Systematic, the Random and the Ugly

Why it is used:
This reference discusses systematic and random components of semiconductor overlay error. It supports treating registration differences as a realistic process variation.

Source:
502_1.tif

Main use in project:
Used to support the overlay perturbation used between structural layers.

O02 - Graff et al. - Robust Control of Maximum Photolithography Overlay Error (2023)

Why it is used:
This recent work provides additional semiconductor-manufacturing context for overlay error and its control. It reinforces the relevance of registration variation in advanced fabrication.

Source:
https://www.sciencedirect.com/science/article/pii/S0007850623000173

Main use in project:
Used as supporting evidence for the overlay/process-variation component.

SEM EDGE BRIGHTENING

E01 - ETH Zürich ScopeM - SEM Imaging with Secondary Electrons

Why it is used:
This SEM imaging reference explains how secondary-electron imaging provides strong surface and edge-related contrast. It supports including enhanced response around feature edges.

Source:
https://scopem.ethz.ch/education/EM-ken/Methods/SEM/SEM_Imaging_1

Main use in project:
Used to justify the secondary-electron edge-contrast component.

E02 - JEOL - Scanning Electron Microscope A to Z

Why it is used:
This practical SEM reference describes secondary-electron imaging and the strong contrast that can occur around edges and protruding features. It supports the qualitative edge response used in the model.

Source:
https://www.jeol.com/applications/pdf/sm/sem_atoz_all.pdf

Main use in project:
Used to support edge and protrusion contrast in the synthetic SEM intensity.

E03 - Scanning Electron Microscopy with Samples in an Electric Field

Why it is used:
This work discusses edge effects and enhanced signal around steep facets and sidewalls in SEM imaging. It provides additional physical support for edge brightening.

Source:
https://pmc.ncbi.nlm.nih.gov/articles/PMC5449051/

Main use in project:
Used to justify the edge-brightening behaviour in the generated search images.